

UNIVERSITY OF SINDH



NAME

KHADIJA INTEKHAB ALAM

ROLL NO

CSE/2K24/80

SUBJECT

HCI AND COMPUTER GRAPHIC

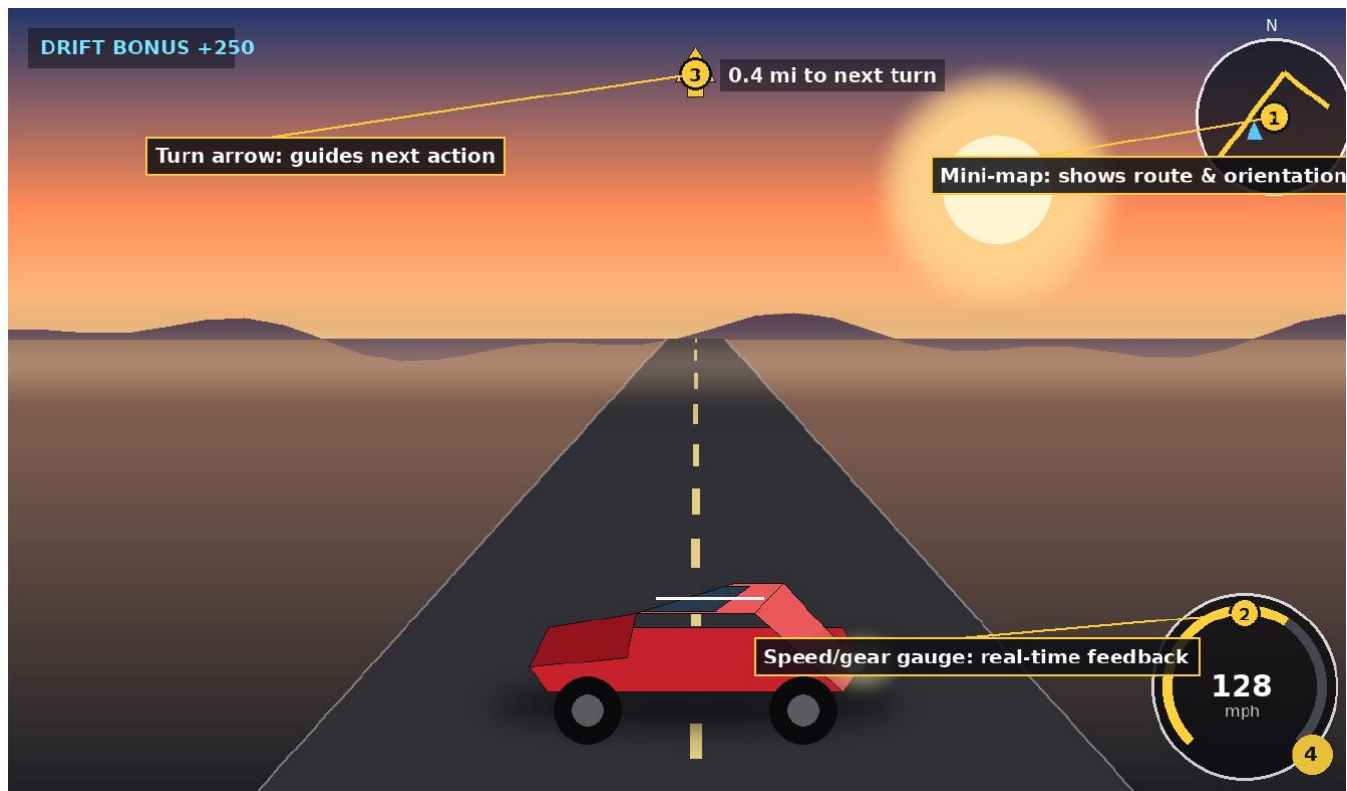
SUBMITTED TO

SIR RAJESH KUMAR

Deconstructing HCI & CG in Daily Software

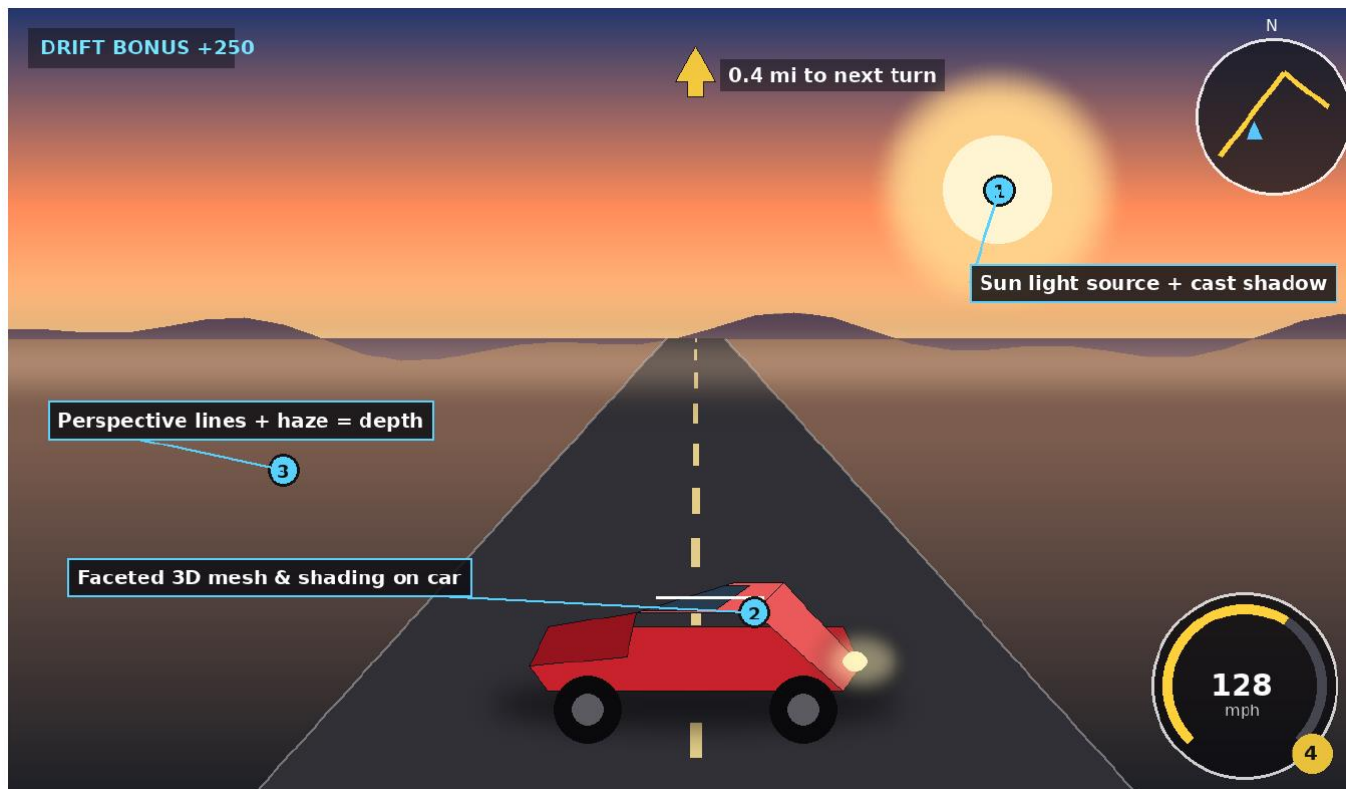
: Visual breakdown of a racing-game interface

Part 1 - HCI Focus: Interaction and UI Elements



Interface element	Why it matters
1. Mini-map (top right)	The mini-map shows where the player is, which direction they are facing, and the road coming up. It helps the player prepare for turns without having to look away from the main road for too long.
2. Speed and gear gauge (bottom right)	This gives constant feedback about the car's current speed and gear . The player can use this information to decide when to brake , accelerate , or change gear .
3. Turn / route arrow (top center)	The route arrow tells the player what direction to take next and how far away the turn is. It is quick to understand and reduces the need to keep checking the full map .

Part 2 - Computer Graphics Focus: Rendering Elements



Interface element	Why it matters
1. Light source and shadow	The sun acts as the main light source . One side of the car appears brighter and a soft shadow appears underneath it. This helps the car feel connected to the road instead of looking flat.
2. 3D mesh and shading	The car is made from simple polygon shapes. Different parts are shaded lighter or darker depending on their angle to the light, which makes the car look three-dimensional .
3. Perspective and atmospheric haze	The road gets narrower as it moves toward the horizon , creating a clear vanishing point . The distant landscape is also lighter and less detailed, which adds a stronger sense of depth .

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Mini Assignment: Visual breakdown of a racing-game interface

Part 3 - Reflection Questions

1. How does the Computer Graphics quality help or hinder the user's ability to interact with the software?

Good **graphics** can make a game easier to understand because the player can judge **distance**, **speed**, and **direction** more quickly. For example, the **road perspective** and the **shading** on the car make the scene feel more **three-dimensional**. However, **graphics** can also become a problem if there is too much **glare**, detail, or **visual clutter**. In that situation, important **HUD** information such as the **mini-map** or **speed gauge** can become harder to notice.